
Forecasting Chaos Across Model Families: An Out-of-Sample, Reliability-Adjusted Benchmark

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Abstract

We study how different families of models forecast a chaotic mechanical system, and how such a comparison should be scored. We construct a controlled k -pendulum benchmark ($k \in \{1, 2, 3\}$; chaotic for $k \geq 2$) and evaluate four families—large language models (LLMs), a time-series foundation model (Chronos), learned dynamics models (Neural ODE, Hamiltonian and Lagrangian neural networks), and classical integrators (Euler, RK4, symplectic)—on the same out-of-sample held-out trajectories, so that learned models are not scored on their training data. Two design choices shape the results: an out-of-sample protocol, and a reliability-adjusted metric that assigns a fixed error to refusals and parse failures rather than discarding them. Out of sample, the learned dynamics models increase their error by approximately a factor of three relative to in-sample evaluation (~ 0.26 to 0.84 – 1.06 rad) and rank below both a zero-shot time-series model and a zero-shot LLM. Among the black-box methods evaluated, the lowest error is obtained by an LLM (kimi, 0.538 rad), while the known-physics integrators, which are given the exact equations, form a ceiling by construction. The training objective affects out-of-sample error (rollout loss is lower than derivative matching, $p_{\text{Holm}} = 1.6 \times 10^{-7}$), but mixed- k rollout, which is best in-sample, shows no significant out-of-sample effect. A matched, confound-free experiment and a direct inference probe find little evidence that the LLMs infer hidden physical constants; under the conditions of our benchmark they default to Earth-standard values. In-sample evaluation can therefore substantially overestimate learned-dynamics performance, which indicates a need for controlled out-of-sample comparison of forecasting methods.

1 Introduction

The outcome of comparing forecasting methods on a chaotic system depends on several choices: whether models are evaluated in or out of sample, how failures are scored, and what information each model receives. We treat these as benchmark-design questions. Current forecasters span several families—LLMs, time-series foundation models, learned dynamics models, and classical integrators—that are seldom evaluated on one system under identical, out-of-sample conditions. Given the state of a dynamical system at time t , we ask which family most accurately predicts the state at $t + T$, and how that comparison should be scored. We use the k -link pendulum: it is integrable at $k=1$ and chaotic with a positive Lyapunov exponent at $k \geq 2$, its ground truth is computable to high precision, and grading requires no domain expertise. All four families are evaluated on this system under identical inputs, horizons, and metrics, with reliability and the structure of chaos treated as part of the evaluation.

Contributions.

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1. A **cross-family evaluation protocol** in which all families are scored on the same out-of-sample held-out trajectories, with a reliability-adjusted metric that assigns a fixed error to abstentions rather than discarding them (§3.4, §5.1).
2. An **out-of-sample leaderboard**: tested off their training trajectories the learned models reach 0.84–1.06 rad (from ~ 0.26 in-sample), and among the models evaluated the lowest black-box error is obtained by an LLM (kimi, 0.538 rad). Rollout loss reduces error relative to derivative matching, while mixed- k rollout shows no significant out-of-sample effect (§5.4).
3. **Chaos-appropriate analysis**: a predictability-horizon metric and a confound-free system-identification test. An unadjusted regime contrast suggests lower error when constants are hidden; a matched paired experiment and a direct inference probe indicate that this is a difficulty confound rather than parameter inference (§5.5).

2 Related Work

Learned dynamics. Neural ODEs learn a continuous-time vector field from data [Chen et al., 2018]. Hamiltonian Neural Networks impose an energy-conserving inductive bias through a learned Hamiltonian [Greydanus et al., 2019]; Lagrangian Neural Networks enforce the Euler–Lagrange equations through second-order autodifferentiation [Cranmer et al., 2020]. Chaotic systems such as the Lorenz attractor and the double pendulum are standard testbeds [Gilpin, 2021]. We evaluate all three against general-purpose models under matched conditions.

Foundation models as forecasters. LLMs can act as zero-shot time-series forecasters when series are serialized as text [Gruver et al., 2023], and time-series foundation models—Chronos [Ansari et al., 2024], TimesFM [Das et al., 2024], and Moirai [Woo et al., 2024]—are trained for general forecasting. Zhang and Gilpin [2025] study zero-shot forecasting of chaotic systems with such models. We place these forecasters on the same grid as physics-informed and language models.

LLMs for physical and mathematical reasoning. LLMs perform well on mathematical problem solving [Lewkowycz et al., 2022] and, with chain-of-thought prompting, on multi-step reasoning [Wei et al., 2022]; recent benchmarks probe physics reasoning in LLMs [Xu et al., 2025]. Our task instead requires quantitative trajectory prediction of a chaotic system rather than qualitative or symbolic reasoning.

3 Benchmark Design

Choice of system. We use the k -link pendulum as a controlled system for evaluating methodology rather than as the object of study. It provides a tunable chaos axis (integrable at $k=1$, chaotic at $k \geq 2$) that a single attractor does not; ground truth to high precision; known governing equations, which provide an oracle ceiling and physics-aware baselines; and controllable constants, which allow parameters to be hidden for the system-identification test (§5.5). A less structured system would confound these effects, so the single controlled system is a deliberate design choice.

3.1 Task and systems

Given the state $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$ of a k -link pendulum at $t=0$, the task is to predict $\mathbf{s}_T$ at horizon T . Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t=10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed. Because the labels are produced by RK4 at this step, the RK4 baseline reported later (the same integrator at a $10\times$ coarser $\Delta t=10^{-2}$) is an oracle ceiling by construction rather than an independent competitor; Euler and the symplectic integrator are distinct known-physics methods. Inputs available to each family are stated in §3.3.

3.2 Regimes and grid

Three regimes test generalization to non-standard constants (Table 1). In `changed_hidden` the constants (g, L, m , damping) are not disclosed to the model, which makes the cell an in-context

Table 1: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
<code>changed_disclosed</code>	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
<code>changed_hidden</code>	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 2: Inputs available to each model family. “Constants” = $(g, L, m, \text{damping})$. Only Chronos receives history; only the learned models receive the true constants in the `hidden` regime.

Family	State $t=0$	History $t<0$	Constants	Equations of motion
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ (all regimes)	— (learned)
Time-series (Chronos)	—	✓ (10 s)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except <code>hidden</code>	—

system-identification problem. An evaluation cell is a tuple (model, k , regime, modality, horizon, prompting); horizons are $\{0.01, 1, 10, 60\}$ s, modalities $\{\text{coords}, \text{image}, \text{image+coords}\}$ (LLMs only), and prompting $\{\text{direct}, \text{chain-of-thought}\}$ (LLMs only). Non-LLM models use the coordinate/direct setting.

Out-of-sample protocol. Learned dynamics models are evaluated off their training data. We generate a shared held-out set (a separate seed from the training trajectories) of 180 initial conditions (60 per regime, spanning $k \in \{1, 2, 3\}$) and evaluate every family on the identical trajectories, keyed by `movement_id`. The cross-family leaderboard (§5.1) uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated (§3.4), giving $180 \times 2 = 360$ cells per model. We report no in-sample numbers and quantify the size of the in-sample effect directly (§5.1).

3.3 Inputs and grouping

Table 2 states what each family receives. Two asymmetries favor the non-LLM models, which makes the comparison conservative for the LLMs: (i) Chronos is the only model that receives the pre- $t=0$ history window; the LLMs and learned models receive only the $t=0$ state. (ii) The learned models receive the true constants in every regime, including `changed_hidden`, because they encode constants as network inputs, whereas the LLMs receive them only when disclosed. Despite these two advantages, the learned models and Chronos rank below an LLM that receives neither (§5.1). A third asymmetry runs in the opposite direction and is larger in magnitude: the LLMs are frontier-scale models with large parameter counts and training budgets, whereas the learned models are small (a 256-wide, 3-layer MLP trained on a single 8 GB GPU; Appendix B). The comparison thus contrasts off-the-shelf frontier models with small learned simulators; the behavior of a substantially larger physics-informed network is outside the scope of this study. We group methods into known-physics (RK4, Euler, symplectic; given the exact equations and constants) and black-box (LLMs, Chronos, learned dynamics). The known-physics methods form a ceiling by construction, and the comparison of interest is among the black-box methods.

3.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories; we also track Cartesian bob error, angular-velocity sign match, and energy drift. Three choices are specific to this setting.

Reliability-adjusted error. A cell is answered only if the model returns a parseable, finite prediction. Reporting error over answered cells alone would reward abstention on hard cells. We therefore assign unanswered cells the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess); we also report the strict upper bound π . Answering is thus part of the score.

Predictability horizon. For chaotic systems, pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ , starting from $t=0$: we walk horizons in increasing order and stop at the first that exceeds τ . The contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. We sweep τ .

System-identification gap. The gap $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$ separates the cost of inferring the constants from being given them. A model that substitutes given numbers is penalized by hiding them ($\Delta > 0$); a model that infers them in context is not ($\Delta \leq 0$). Measuring this requires holding the trajectory fixed and toggling only disclosure; otherwise disclosure is confounded with each regime’s constants and initial conditions. We therefore run a matched experiment (§5.5): 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, evaluated under disclosed and hidden conditions on identical trajectories, with a probe that asks the hidden-condition model to report its inferred constants.

4 Methods

LLMs. We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured prompt (system description, disclosed constants where applicable, initial state) and a JSON answer schema (Appendix A). We use a uniform configuration (65,536 max tokens, 300 s timeout) for all three; reasoning traces are read from the dedicated reasoning channel when present. We take a single deterministic sample per cell (temperature 0); no best-of- N aggregation is used, so reliability reflects the raw per-call behavior. The API cost of the out-of-sample LLM evaluation (Parts A and B, 3,960 cells) was \$110.

Time-series foundation models. We run Chronos [Ansari et al., 2024] locally on a single GPU. Chronos is the only model given history: its adapter takes the 10 s pre- $t=0$ window of all $2k$ state channels, resamples it onto a uniform grid of up to 512 points, and forecasts to the target horizon (prediction length ≤ 64 steps). The leaderboard uses `chronos-2`, which forecasts either each channel independently (univariate) or all $2k$ channels jointly (multivariate); this tests, with one model, whether modeling cross-channel coupling helps on a coupled system. For that test we use 20 trajectories per (k , regime) cell and compare the two modes paired on identical trajectories (Wilcoxon signed-rank, bootstrap confidence intervals, Holm correction). The earlier univariate `chronos-t5-large` was used only in a preliminary round; TimesFM is univariate; Moirai’s `uni2ts` could not be installed in our environment.

Learned dynamics. Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and integrate at inference with `scipy`. We compare two objectives: derivative matching (MSE on finite-difference $\dot{s}$) and rollout loss (MSE on the state after k Euler steps, with gradients through the unroll). We also use mixed- k rollout, which trains on several window lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation, which Apple’s MPS backend does not run; we train it at full scale (67k samples, hidden 256) on a CUDA GPU.

Integrators. Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations of motion; their error is from discretization alone. RK4 is the data-generating process and serves as an oracle ceiling rather than an independent competitor.

5 Results

5.1 Out-of-sample leaderboard

Table 3 reports the out-of-sample leaderboard on the shared held-out set (360 cells: 180 trajectories $\times$ horizons $\{1, 10\}$ s), using the reliability-adjusted metric so that an unanswered or missing cell is scored at $\pi/2$. Every model is scored on the same grid; the Answered column gives the fraction of cells for which a model returned a valid prediction (only the LNN, whose autograd-ODE integration

Table 3: Out-of-sample leaderboard on the shared held-out set (360 cells, horizons $\{1, 10\}$ s). Reliability-adjusted mean angle error (rad): unanswered or missing cells scored at $\pi/2$; 95% bootstrap CI. RK4 is the data-generating process, hence an oracle ceiling rather than a competitor. Methods new in this work are marked $\dagger$.

#	Model	Err	95% CI	Answered	Family
–	rk4 (oracle)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi $\dagger$	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni $\dagger$	0.697	[0.631, 0.766]	100%	time-series
6	neural-ode-rollout-mixed $\dagger$	0.844	[0.770, 0.920]	100%	learned
7	neural-ode-rollout	0.870	[0.795, 0.947]	100%	learned
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	hnn-rollout	0.922	[0.847, 0.998]	100%	learned
10	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
11	hnn-rollout-mixed $\dagger$	1.003	[0.923, 1.084]	100%	learned
12	neural-ode	1.028	[0.953, 1.107]	100%	learned
13	hnn	1.064	[0.982, 1.145]	100%	learned
14	l1nn $\dagger$ (full-scale)	1.421	[1.363, 1.481]	45%	learned

diverges to NaN on 55% of cells, falls below 100%). LLMs are scored on their direct-prompt forecasts. Confidence intervals are 95% paired bootstrap ($B=10^4$).

The classical integrators have the lowest error: RK4 (the data-generating process, hence an oracle ceiling) at 0.059, symplectic at 0.205, Euler at 0.405. Among the black-box methods evaluated, the lowest error is obtained by kimi (0.538 rad, 99% answered), below both Chronos modes and all learned dynamics models. The learned models, which score ~ 0.26 rad under in-sample evaluation, reach 0.84–1.06 rad out of sample; the lowest (mixed- k rollout Neural ODE, 0.844) ranks 6th, below the integrators, kimi, and both Chronos modes. This factor of approximately three between in-sample and out-of-sample error indicates that the in-sample strength of small learned dynamics models on this benchmark is largely attributable to evaluation on their training trajectories.

Effect of the reliability adjustment. grok answers 82% of the grid and l1nn 45%; scoring those abstentions at $\pi/2$, rather than averaging over answered cells, places them at ranks 8 and 14. Averaging over answered cells alone would rank the least reliable models higher.

5.2 Behavior by horizon

Table 4 reports out-of-sample error by horizon. At 1 s the ordering follows the leaderboard: the integrators are near-exact, kimi (0.457) has the lowest black-box error, and the learned models are at 0.76–0.97. At 10 s all models are in the chaotic regime, and the ordering is stable: kimi (0.619) remains lowest among black-box methods while the learned models reach 0.91–1.26. The mixed- k Neural ODE, which is the in-sample standout at 1–10 s, is mid-pack here (0.779 / 0.908). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the leaderboard is the range over which the families remain separable.

5.3 Breakdown by k and by regime

Tables 8 and 9 (Appendix C) report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for kimi, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The changed_hidden regime has the lowest error for almost every model (for kimi, 0.46 hidden vs. 0.64 normal; the learned models are lower by 0.5–0.8 rad relative to normal); its low-gravity, short-rod constants produce slower, less-divergent motion over a fixed horizon, so the unadjusted “hidden helps” contrast is a difficulty confound (§5.5). Re-estimating the leaderboard confidence intervals with a cluster bootstrap over whole held-out trajectories changes no interval by more than

Table 4: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (oracle)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
neural-ode-rollout-mixed	0.779	0.908
neural-ode-rollout	0.790	0.950
hnn-rollout	0.758	1.085
grok-4-1-fast-reasoning	0.905	0.857
neural-ode	0.793	1.264
deepseek-v4-pro	0.996	0.909
lnn (full-scale)	1.335	1.508

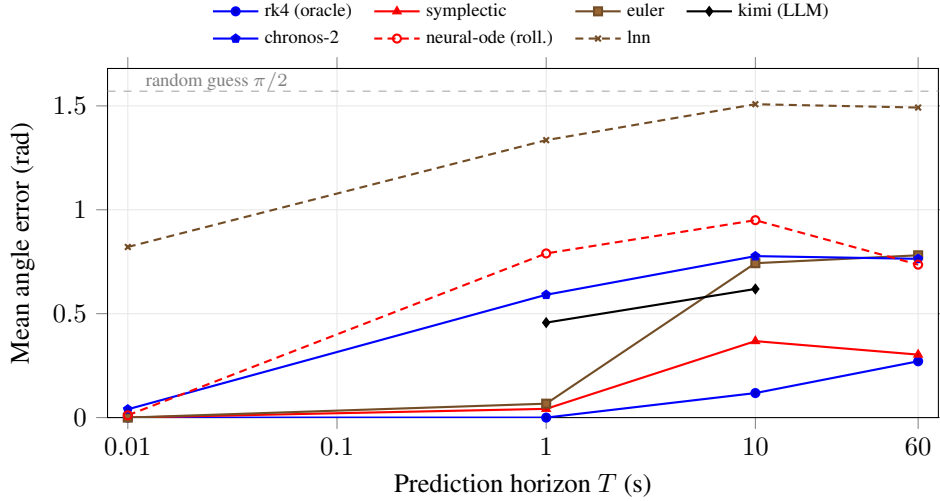


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s. Dashed line: random-guess baseline $\pi/2$.

~ 0.03 rad and changes no ranking, so the gaps are not an artifact of treating correlated cells as independent.

5.4 Training objective

Table 5 reports the effect of the loss with architecture fixed. Rollout loss has lower error than derivative matching: for the Neural ODE, 1.028 drops to 0.870 under a single- k rollout loss, and in a powered paired study ($n=720$ trajectories, Wilcoxon signed-rank) this difference is significant ($p_{\text{Holm}}=1.6 \times 10^{-7}$). Penalizing trajectory error through an unrolled integrator, rather than instantaneous derivative error, optimizes the quantity the benchmark measures. The in-sample advantage of mixed- k rollout does not hold out of sample: for the Neural ODE it is indistinguishable from single- k (0.844 vs. 0.870; $p_{\text{Holm}}=0.089$), and for the HNN it is higher (1.003 vs. 0.922; $p_{\text{Holm}}=0.088$). Full-scale CUDA training of the LNN gives the highest error (1.421), with 55% of cells diverging to NaN under autograd-ODE integration. We do not claim that LNNs are intrinsically weak; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Table 5: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.028	—
	rollout (single- k)	0.870	sig. ($p_{\text{Holm}}=1.6 \times 10^{-7}$)
	mixed- k	0.844	n.s. ($p_{\text{Holm}}=0.089$)
HNN	derivative MSE	1.064	—
	rollout (single- k)	0.922	(rollout lower)
	mixed- k	1.003	n.s. ($p_{\text{Holm}}=0.088$)

Table 6: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better.

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (oracle)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	60
neural-ode-rollout-mixed	0.01	0.01	0.01	60
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	1.0
lnn	0.01	0.01	0.01	0.01

5.5 Chaos-appropriate analysis

Predictability horizon. Table 6 reports the longest horizon each model tracks contiguously below a threshold τ , over the ladder $\{0.01, 1, 10, 60\}$ s available for the non-LLM models. Only the numerical integrators stay below strict thresholds past 1 s: RK4 tracks to 10 s at $\tau=0.25$ and 60 s at $\tau=0.5$, symplectic to 60 s at $\tau=0.5$, Euler to 60 s only at $\tau=1.0$. Every learned model, including the rollout and mixed- k Neural ODEs, returns to the first rung (0.01 s) at all strict thresholds and reaches 60 s only at $\tau=1.0$ ($\sim 57^\circ$), where the threshold is uninformative. The LLMs were evaluated only at $\{1, 10\}$ s, so their ladder is coarser; within it, only kimi is below 0.5 rad at 1 s. Within our thresholds, no data-driven model exceeds the predictability horizon of the numerical integrators.

Unadjusted regime contrast. On the shared held-out set the three regimes differ in both their constants and their initial conditions, so a regime contrast conflates disclosure with difficulty. The unadjusted contrast is large: pooled across LLMs, the hidden regime has lower error than the disclosed regime (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and than the normal regime (-0.524 , $p_{\text{Holm}} < 10^{-3}$). This is the “LLMs do better when constants are hidden” pattern reported in earlier rounds.

Matched experiment. We ran the confound-free test of §3.4: 5 non-standard constant sets, 10 trajectories each, $k \in \{1, 2, 3\}$, scored under disclosed and hidden conditions on identical trajectories, paired by trajectory (Table 7). The effect reverses. Hiding the constants increases error ($\Delta_{\text{sysid}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$); at $k=1$ disclosure helps ($+0.297$, $p_{\text{Holm}} < 10^{-3}$) and at the 1 s horizon as well ($+0.145$, $p_{\text{Holm}} < 10^{-3}$), with a small reverse effect at $k=3$ (-0.055 , $p_{\text{Holm}} = 0.016$). Per model, kimi is penalized by hiding ($\Delta = +0.211$), grok mildly ($+0.038$), and deepseek is flat (-0.042). The advantage of the hidden regime in the unadjusted contrast is therefore a difficulty confound: the hidden-regime trajectories are easier.

Inference probe. In the hidden condition we also asked each model to report its inferred constants. The models do not recover them: against true $g \in [1.6, 12]$ the inferred- g MAE is 1.81 (deepseek), 1.88 (grok), and 2.12 (kimi). The pattern is consistent across models: they default to Earth-standard values (deepseek reports $g \approx 9.81$ on nearly every cell; grok and kimi report $L=m=1.0$ on nearly

Table 7: Matched, confound-free system-identification gap (rad), paired on identical trajectories. $\Delta_{\text{sysid}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected.

Model	disclosed	hidden	Δ_{sysid}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

every cell, a constant output for which the rank correlation with truth is undefined). The one nonzero signal is deepseek’s rod lengths, which correlate with truth at Spearman 0.11 ($p=0.007$). We therefore find little evidence that the accuracy difference between regimes is caused by in-context system identification.

5.6 Univariate versus multivariate time-series modeling

The leaderboard uses one trajectory per cell, which resolves the large gaps between families but not small effects. To test whether a time-series model that shares information across the $2k$ state channels benefits from their coupling, we compare Chronos-2 in univariate and multivariate modes with 20 trajectories per (k , regime) cell ($n=720$ paired samples) on identical trajectories. The two modes are not distinguishable: mean error 0.549 (uni, 95% CI [0.505, 0.593]) vs. 0.543 (multi, [0.498, 0.589]). The paired difference is +0.005 rad overall, with no significant effect at any k or at horizons 1 s and 10 s; the per- k trend at $n=1$ (multivariate appearing to help most at $k=3$) goes to zero at $n=720$. A nominal overall $p=0.025$ is driven by a sub-0.01 rad bias at the 0.01 s horizon and does not survive Holm correction. We report a null result: cross-channel coupling gives no measurable benefit for time-series forecasting of this system. On the shared out-of-sample grid the Chronos family is at 0.68–0.70 rad, below every learned model and above only the numerical integrators and kimi (Table 3).

6 Discussion

In-sample evaluation. When the learned dynamics models are scored on the same held-out trajectories as the other families, rather than on their training data, their error increases by approximately a factor of three ($\sim 0.26 \rightarrow 0.84\text{--}1.06$ rad) and falls below a zero-shot time-series model and a zero-shot LLM. The conclusion that a small fitted model outperforms general-purpose forecasters, which an in-sample leaderboard supports, does not hold out of sample. For chaotic systems, a learned simulator should be reported on trajectories it was not trained on. Out of sample, the lowest black-box error in our benchmark is obtained by an LLM (kimi); among the learned models, what transfers is the rollout objective rather than the mixed- k variant.

Reliability. Averaging over answered cells alone ranks the least reliable models higher. Scoring abstentions at $\pi/2$ places grok (82% answered) at rank 8 and the LNN (45%) last. Chain-of-thought does not change this: paired CoT-vs-direct differences are not significant for any LLM (kimi is slightly higher with CoT). A fixed budget of reasoning tokens does not substitute for the sequential integration the task requires.

System identification. The “LLMs do better when constants are hidden” effect is a difficulty confound. In a matched experiment that toggles only disclosure on identical trajectories, hiding the constants increases error (+0.069 rad overall; +0.297 at $k=1$), and the inference probe shows the models do not recover the constants (g MAE $\sim 1.8\text{--}2.1$ against true $g \in [1.6, 12]$); they default to Earth-standard values. Under the conditions of our benchmark we find little evidence that the models infer hidden parameters, and regime contrasts should not be read as inference without a matched control.

Cost. The out-of-sample LLM evaluation cost \$110 in API fees (3,960 cells, $\sim$ \$0.03 per forecast) and is the only family billed per query; the numerical and time-series models are free at inference, and the learned models are free after a one-time training cost (minutes to a few hours on one 8 GB GPU). The lowest-error black-box method is also the most expensive, while a model trained on the system runs at 100% reliability and modestly higher error.

Scope. The quantitative rankings are specific to this system and horizon; we do not claim that a particular model order transfers to Lorenz, driven-pendulum, or fluid systems. The protocol does transfer: in-sample scoring of a fitted simulator is optimistic on any system whose train and test trajectories overlap, and the reliability-adjusted metric and the matched disclosure control are system-agnostic. We expect the resulting ranking to be system-dependent, since which models abstain or destabilize, and hence how the reliability adjustment reorders them, may differ on another system. The harness is substrate-parametrized (the dynamics enter only through the ground-truth integrator and the known equations), so applying it to other chaotic systems is a direct extension.

7 Limitations

The leaderboard aggregates 180 held-out trajectories (60 per regime) with bootstrap confidence intervals, and the powered sub-studies (objective, multivariate, system-ID) use $n=720\text{--}900$ paired samples; all trajectories derive from a single held-out seed, and additional seeds would tighten the error bars. Predictions are single-endpoint rather than full-trajectory; a full-rollout protocol would allow the predictability horizon to be measured directly rather than inferred from four horizons. The comparison is structurally asymmetric: the learned models are trained on the k -pendulum family and tested on held-out trajectories of it, whereas the LLMs and Chronos are zero-shot. The pendulum equations are common in training corpora, so contamination cannot be excluded; the hidden-constants regime mitigates this only in part. Finally, the numerical results are demonstrated on a single system and three LLMs; extending them to other chaotic systems (Lorenz, driven pendulum) and a broader model set (e.g. Qwen-VL, TimesFM, Moirai) through the same harness is left to future work.

8 Conclusion

On a controlled k -pendulum benchmark spanning four model families, evaluated on the same out-of-sample held-out trajectories, in-sample evaluation of learned dynamics models is optimistic: scored off their training data the learned models increase their error by approximately a factor of three and rank below both a zero-shot time-series model and a zero-shot LLM, so that the lowest black-box error in our benchmark is obtained by an LLM (kimi), while the numerical integrators form the ceiling when the equations are known. The objective, not the architecture, transfers (rollout loss is lower than derivative matching; mixed- k shows no out-of-sample effect). Reliability-adjusted scoring and chaos-appropriate metrics change the leaderboard relative to a naive ranking, and a matched experiment with a direct inference probe finds little evidence of in-context system identification.

Reproducibility. Code, prompts (Appendix A), trained weights, the shared held-out specification, and per-cell checkpoints are released with the benchmark. Training trajectories use seed 42; the held-out evaluation set uses a disjoint seed (20260620), 180 trajectories (60 per regime). LLMs (kimi-k2.6, grok-4-1-fast-reasoning, deepseek-v4-pro, via Azure AI Foundry, evaluated June 2026) are sampled once at temperature 0. A cell is answered if it returns a parseable, finite prediction; otherwise it is assigned $\pi/2$. All local training and evaluation run on a single consumer GPU (8 GB); no proprietary training data is used.

Author Contributions

S.N.K. (lead author, corresponding) designed the benchmark, led the evaluation harness and the learned-dynamics, time-series, and analysis pipelines, ran the local out-of-sample and significance experiments, and wrote the manuscript. T.S. helped plan and build the benchmark harness. S.S. ran the large language model (API) evaluations. Author order reflects relative contribution. All authors reviewed and approved the final manuscript.

References

- Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, et al. Chronos: Learning the language of time series. *Transactions on Machine Learning Research (TMLR)*; arXiv:2403.07815, 2024.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- Miles Cranmer, Sam Greydanus, Stephan Hoyer, Peter Battaglia, David Spergel, and Shirley Ho. Lagrangian neural networks. *ICLR 2020 Workshop on Integration of Deep Neural Models and Differential Equations*; arXiv:2003.04630, 2020.
- Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2310.10688.
- William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2110.05266.
- Sam Greydanus, Misko Dzamba, and Jason Yosinski. Hamiltonian neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- Nate Gruver, Marc Finzi, Shikai Qiu, and Andrew Gordon Wilson. Large language models are zero-shot time series forecasters. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. arXiv:2310.07820.
- Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, et al. Solving quantitative reasoning problems with language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2206.14858.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. arXiv:2201.11903.
- Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.02592.
- Xin Xu, Qiyun Xu, Tong Xiao, Tianhao Chen, Yuchen Yan, Jiaxin Zhang, Shizhe Diao, Can Yang, and Yang Wang. Ugphysics: A comprehensive benchmark for undergraduate physics reasoning with large language models. In *International Conference on Machine Learning (ICML)*, 2025. arXiv:2502.00334.
- Yuanzhao Zhang and William Gilpin. Zero-shot forecasting of chaotic systems. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2409.15771.

A Prompt templates

The prompts (from `bench/prompts.py`) are reproduced below.

System prompt (direct / no-CoT).

You are a careful physicist predicting the future state of a k-pendulum (a chain of k point masses on massless rods, hinged in a planar chain, swinging under gravity with viscous damping). Respond with ONLY a single JSON object on one line, no prose, no code fences. Schema: {"theta": [...k floats..], "omega": [...k floats..]} where theta_i is in radians (link i measured from straight-down, positive counter-clockwise) and omega_i is angular velocity in rad/s.

System prompt (chain-of-thought). Same preamble, then: “Reason step-by-step about the dynamics, then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a single JSON object: {"theta": [...k floats..], "omega": [...k floats..]}.”

Table 8: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (oracle)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
neural-ode-rollout-mixed	0.616	0.868	1.047
neural-ode-rollout	0.661	0.834	1.116
grok-4-1-fast-reasoning	0.366	1.030	1.247
hnn-rollout	0.508	0.921	1.336
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout-mixed	0.516	1.083	1.410
neural-ode	0.509	1.167	1.409
hnn	0.491	1.230	1.470
lnn	1.146	1.547	1.571

User prompt (coordinate modality).

Number of links $k = \{k\}$
Physical constants:
 g (gravity, m/s^2): $\{g\}$
 L (rod lengths, m): $\{L\}$
 m (point masses, kg): $\{m\}$
 damping (viscous coefficient on ω): $\{\text{damping}\}$
Initial state at $t=0$:
 θ_0 (rad): $\{\theta_0\}$
 ω_0 (rad/s): $\{\omega_0\}$
Predict the state at $t = \{T\}$ seconds.
Return θ and ω as length- $\{k\}$ JSON arrays.

Hidden-constants regime. The constants block above is replaced by:

Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point masses, and damping are NOT disclosed. You must infer dynamics from the initial conditions alone.

Parsing and scoring. Each response is stripped of code fences and (for CoT) of everything outside the `<answer>...</answer>` tags; the first `\{...\}` object is JSON-parsed and validated to contain length- k θ and ω arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

B Training and compute

All learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, on a single NVIDIA GTX 1070 Ti (8 GB). Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at $\Delta t = 0.01$ s from the 10^{-3} s ground truth.

C Per- k and per-regime breakdowns

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 8) and by regime (Table 9); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$, so each row is a mean over the same grid as Table 3.

Table 9: Out-of-sample error by regime. `changed_hidden` has the lowest error for almost every model; its low-gravity/short-rod constants give slower, less-divergent motion, which is why the unadjusted regime contrast (§5.5) is a difficulty confound rather than evidence of in-context system identification.

Model	normal	disclosed	hidden
rk4 (oracle)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimik2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
neural-ode-rollout-mixed	1.260	0.836	0.434
neural-ode-rollout	1.281	0.894	0.436
grok-4-1-fast-reasoning	1.096	0.959	0.588
hnn-rollout	1.087	1.115	0.564
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout-mixed	1.244	1.172	0.592
neural-ode	1.221	1.095	0.770
hnn	1.262	1.303	0.626
lhn	1.511	1.408	1.345